



Netflix Movies & TV Shows - Big Data Analytics Project



Project Overview

This project performs comprehensive end-to-end analysis of the Netflix Movies & TV Shows dataset using Python, SQL, and Power BI. The analysis covers data cleaning, exploratory data analysis, statistical insights, and interactive dashboard creation to derive actionable business insights.



Repository Structure

```
netflix-analytics/
├── README.md           # Project documentation
├── data/
│   ├── netflix_titles.csv    # Raw dataset
│   └── netflix_cleaned.csv   # Cleaned dataset
├── notebooks/
│   └── netflix_analysis.ipynb # Python analysis notebook
├── sql/
│   └── queries.sql          # SQL queries
├── powerbi/
│   └── netflix_dashboard.pbix # Power BI dashboard
├── report/
│   └── project_report.pdf    # Detailed project report
└── presentation/
    └── slides.pptx          # Presentation slides
```



Dataset Information

Source: Netflix Movies and TV Shows Dataset (Kaggle)

Description: Contains information about movies and TV shows available on Netflix, including:

- Show ID, Title, Type (Movie/TV Show)
- Director, Cast, Country
- Date Added, Release Year
- Rating, Duration
- Listed In (Genres), Description

Dataset Size: ~8,800+ titles

Technologies Used

- **Python 3.x:** Data cleaning, analysis, and visualization
 - pandas, numpy
 - matplotlib, seaborn
 - **SQL:** Database queries and data manipulation
 - SQLite / MySQL / PostgreSQL
 - **Power BI:** Interactive dashboard and business intelligence
 - **Jupyter Notebook:** Interactive development environment
 - **Git & GitHub:** Version control and collaboration
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Analysis Components

1. Python Analysis (notebooks/netflix_analysis.ipynb)

Key Tasks:

- Data loading and initial exploration
- Data quality assessment (missing values, duplicates)
- Data cleaning and preprocessing
- Exploratory Data Analysis (EDA)
- Statistical analysis
- Data visualizations

Key Insights Derived:

- Content type distribution (Movies vs TV Shows)
- Geographic distribution of content
- Temporal trends in content addition
- Genre popularity analysis

- Rating distribution
- Duration analysis for movies and seasons for TV shows

2. SQL Analysis (sql/queries.sql)

Query Categories:

- Basic data exploration
- Content type analysis
- Country-wise production analysis
- Temporal analysis (yearly/monthly trends)
- Genre and category analysis
- Rating distribution
- Director analysis
- Advanced business queries

Key SQL Queries:

- Top producing countries
- Year-over-year growth analysis
- Content concentration analysis
- Time gap between release and Netflix addition
- Rating trends over time

3. Power BI Dashboard (powerbi/netflix_dashboard.pbix)

Dashboard Features:

- Interactive filters by year, country, type, and rating
- KPI cards showing total content, movies, TV shows
- Geographic map visualization
- Time series charts for trends
- Genre distribution charts
- Top countries and directors visuals

How to Run This Project

Prerequisites

```
bash

# Python libraries
pip install pandas numpy matplotlib seaborn jupyter

# SQL database (choose one)
# SQLite (built-in with Python)
# MySQL
# PostgreSQL
```

Step 1: Clone the Repository

```
bash

git clone https://github.com/yourusername/netflix-analytics.git
cd netflix-analytics
```

Step 2: Download the Dataset

1. Download the Netflix dataset from [Kaggle](#)
2. Place `netflix_titles.csv` in the `data/` folder

Step 3: Run Python Analysis

```
bash

# Open Jupyter Notebook
jupyter notebook

# Navigate to notebooks/netflix_analysis.ipynb
# Run all cells
```

The notebook will:

- Clean the data
- Generate visualizations

- Export `netflix_cleaned.csv` to the data folder

Step 4: Run SQL Queries

```
bash

# For SQLite
sqlite3 netflix.db

# Import cleaned data
.mode csv
.import data/netflix_cleaned.csv netflix_data

# Run queries from sql/queries.sql
.read sql/queries.sql
```

For MySQL/PostgreSQL, import the CSV and execute the queries using your preferred tool.

Step 5: Open Power BI Dashboard

1. Open Power BI Desktop
2. Open `powerbi/netflix_dashboard.pbix`
3. Update data source to point to your `netflix_cleaned.csv`
4. Refresh the dashboard

Key Findings

Content Distribution

- **Movies:** ~70% of total content
- **TV Shows:** ~30% of total content

Geographic Insights

- **Top Producer:** United States (highest content volume)
- **Emerging Markets:** India, UK, Japan showing growth

Temporal Trends

- Significant increase in content addition from 2015-2020
- Peak addition months: October, November, December

Genre Insights

- Most popular genres: International Movies, Dramas, Comedies
- Documentaries and Stand-up Comedy growing in recent years

Rating Analysis

- Most common rating: TV-MA (Mature Audience)
- Family-friendly content (TV-G, PG) represents ~15%

Content Strategy

- Average gap between release and Netflix addition: 2-3 years
 - Trend towards acquiring recent releases
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Project Report

Detailed analysis, methodology, and insights are documented in:

- **File:** `report/project_report.pdf`
 - **Presentation:** `presentation/slides.pptx`
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Learning Outcomes

Through this project, I demonstrated:

-  Data cleaning and preprocessing techniques
-  Exploratory Data Analysis (EDA)
-  Statistical analysis and visualization
-  SQL database querying and data manipulation

- ☒ Business Intelligence dashboard creation
 - ☒ Deriving actionable insights from data
 - ☒ Technical documentation and presentation
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Future Enhancements

- ☐ Sentiment analysis on descriptions
 - ☐ Recommendation system development
 - ☐ Predictive modeling for content success
 - ☐ Integration with external APIs for real-time data
 - ☐ Advanced NLP for genre classification
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Author

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License

This project is open source and available under the MIT License.

Acknowledgments

- Dataset: Kaggle Netflix Movies and TV Shows Dataset
 - Inspiration: Big Data Analytics Course Assignment
 - Tools: Python, SQL, Power BI
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Contact

For questions or feedback, please open an issue or reach out via email.

★ If you found this project helpful, please give it a star!